=== src/m3/answers_3.html === <section class="page"> Section B · Paragraph / problem answers (Part 2: Q29–Q36)<small>6 marks each. Ceiling 72.</small>
29PYQ MTS-26 Q8 State and prove the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1). Know firstT20T03T18P12M05 Theorem 7.3.1 · Fundamental Theorem of Calculus, First Form (Bartle p. 216)Suppose there is a finite set E in [a,b] and functions f, F : [a,b] \to \mathbb{R} such that: (a) F is continuous on [a,b]; (b) F'(x) = f(x) for all x \in [a,b] \setminus E; (c) f belongs to \mathcal{R}[a,b]. Then we have
\int_a^b f = F(b) - F(a).
 SkeletonFor E = \{a,b\}, given \varepsilon > 0 choose \delta_\varepsilon from f \in \mathcal{R}[a,b] and a partition P with \lVert P\rVert < \delta_\varepsilon. Apply the MVT 6.2.4 to F on each [x_{i-1},x_i] to get F(x_i) - F(x_{i-1}) = f(u_i)(x_i - x_{i-1}); sum to get F(b) - F(a) = S(f;\dot P_u), which is within \varepsilon of \int_a^b f. Proof of Theorem 7.3.1 (Bartle pp. 216–217) We first prove the case where E = \{a,b\}. Let \varepsilon > 0 be given. Since f \in \mathcal{R}[a,b] by hypothesis (c), there exists \delta_\varepsilon > 0 such that if \dot P is any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, then
\left\lvert S(f;\dot P) - \int_a^b f\right\rvert < \varepsilon.
Definition 7.1.1 (T03). Let P = \{[x_{i-1},x_i]\}_{i=1}^n be a partition of [a,b] with \lVert P\rVert < \delta_\varepsilon. Since F is continuous on [x_{i-1},x_i] by (a) and differentiable on (x_{i-1},x_i) \subseteq [a,b]\setminus E by (b), the Mean Value Theorem 6.2.4 (P12) applied to F on [x_{i-1},x_i] implies there exists u_i \in (x_{i-1},x_i) such that
F(x_i) - F(x_{i-1}) = F'(u_i)(x_i - x_{i-1}) = f(u_i)(x_i - x_{i-1}) \quad \text{for } i = 1, \dots, n.
Mean Value Theorem 6.2.4 on [x_{i-1},x_i], using u_i \notin E \Rightarrow F'(u_i) = f(u_i). Adding these n equations and telescoping the left-hand side gives
F(b) - F(a) = \sum_{i=1}^n \big(F(x_i) - F(x_{i-1})\big) = \sum_{i=1}^n f(u_i)(x_i - x_{i-1}) = S(f;\dot P_u),
where \dot P_u := \{([x_{i-1},x_i],u_i)\}_{i=1}^n is a tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon.Telescoping sum (M01, M05). Substituting S(f;\dot P_u) = F(b) - F(a) into Step 1 gives \big\lvert F(b) - F(a) - \int_a^b f\big\rvert < \varepsilon. Since \varepsilon > 0 is arbitrary, \int_a^b f = F(b) - F(a).Arbitrary \varepsilon rule (P05). General finite set E (Exercise 7.3.1): Let a = c_0 < c_1 < \cdots < c_m = b be the points of E \cup \{a,b\} in increasing order. On each [c_{k-1},c_k], F is continuous, F'(x) = f(x) on (c_{k-1},c_k), and f|_{[c_{k-1},c_k]} \in \mathcal{R}[c_{k-1},c_k] by Corollary 7.2.10. Applying the first part on each subinterval and summing via Corollary 7.2.11 gives \int_a^b f = \sum_{k=1}^m \big(F(c_k) - F(c_{k-1})\big) = F(b) - F(a). ∎Corollaries 7.2.10 and 7.2.11 (T18).   Marks: statement with (a),(b),(c) 1.5 · MVT step on [x_{i-1},x_i] 2 · telescoping to S(f;\dot P_u) and conclusion 1.5 · general finite E 1. </section>

<section class="page"> 30PYQ MTS-25 Q6also MAT-22 Q18, MAT-21 Q19 Substitution Theorem 7.3.8: statement, proof, five substitution evaluations, and \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt via FTC 1. Know firstT25T23T20T18P11 Theorem 7.3.8 · Substitution Theorem (Bartle p. 220)Let J := [\alpha,\beta] and let \varphi : J \to \mathbb{R} have a continuous derivative on J. If f : I \to \mathbb{R} is continuous on an interval I containing \varphi(J), then
\int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx.
 Proof of Theorem 7.3.8 (Bartle Exercise 7.3.17, p. 224) Define F(u) := \int_{\varphi(\alpha)}^u f(x)\,dx for u \in I. Since f is continuous on I, Theorem 7.3.6 (FTC 2) implies that F'(u) = f(u) for all u \in I.Theorem 7.3.6 (T23). Define H(t) := F(\varphi(t)) for t \in J = [\alpha,\beta]. By the Chain Rule 6.1.6, H is differentiable on J and H'(t) = F'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t) for all t \in J.Chain Rule 6.1.6 (P11). Since f, \varphi, and \varphi' are continuous, (f\circ\varphi)\varphi' is continuous on [\alpha,\beta] and hence in \mathcal{R}[\alpha,\beta] (Theorem 7.2.7). Since H(\alpha) = F(\varphi(\alpha)) = 0, Theorem 7.3.1 (FTC 1 with E = \varnothing) gives
\int_\alpha^\beta f(\varphi(t))\,\varphi'(t)\,dt = H(\beta) - H(\alpha) = F(\varphi(\beta)) = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x)\,dx. \quad \blacksquare
Theorem 7.2.7 and Theorem 7.3.1 (T20).
Five substitution evaluations (Example 7.3.9(a) & Exercise 7.3.18):
1. \int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt (MTS-25 Q6, MAT-21 Q19): Put \varphi(t) = \sqrt t on [1,4] (\varphi'(t) = \frac{1}{2\sqrt t} continuous on [1,4]) and f(x) = 2\sin x:
\int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = \int_1^2 2\sin x\,dx = \big[-2\cos x\big]_1^2 = 2(\cos 1 - \cos 2) \approx 1.912898.
 2. \int_1^4 \frac{\cos\sqrt t}{\sqrt t}\,dt (Ex 7.3.18(d)): Put \varphi(t) = \sqrt t on [1,4] and f(x) = 2\cos x:
\int_1^4 \frac{\cos\sqrt t}{\sqrt t}\,dt = \int_1^2 2\cos x\,dx = \big[2\sin x\big]_1^2 = 2(\sin 2 - \sin 1) \approx 0.135653.
 3. \int_0^1 t\sqrt{1+t^2}\,dt (Ex 7.3.18(a)): Put \varphi(t) = 1 + t^2 on [0,1] (\varphi'(t) = 2t continuous) and f(x) = \tfrac12\sqrt x:
\int_0^1 t\sqrt{1+t^2}\,dt = \int_1^2 \frac12 x^{1/2}\,dx = \left[\frac13 x^{3/2}\right]_1^2 = \frac{2\sqrt 2 - 1}{3} \approx 0.609476.
 4. \int_0^2 t^2(1+t^3)^{-1/2}\,dt (Ex 7.3.18(b)): Put \varphi(t) = 1 + t^3 on [0,2] (\varphi'(t) = 3t^2 continuous) and f(x) = \tfrac13 x^{-1/2}:
\int_0^2 t^2(1+t^3)^{-1/2}\,dt = \int_1^9 \frac13 x^{-1/2}\,dx = \left[\frac23\sqrt x\right]_1^9 = \frac23(3 - 1) = \frac{4}{3}.
 5. \int_1^4 \frac{\sqrt{1+\sqrt t}}{\sqrt t}\,dt (Ex 7.3.18(c)): Put \varphi(t) = 1 + \sqrt t on [1,4] (\varphi'(t) = \frac{1}{2\sqrt t} continuous on [1,4]) and f(x) = 2\sqrt x:
\int_1^4 \frac{\sqrt{1+\sqrt t}}{\sqrt t}\,dt = \int_2^3 2x^{1/2}\,dx = \left[\frac43 x^{3/2}\right]_2^3 = \frac43(3\sqrt 3 - 2\sqrt 2) \approx 3.156967.
  Why 7.3.8 fails on [0,4] and how \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt is evaluated (Example 7.3.9(b); PYQ MAT-22 Q18): On [0,4], \varphi(t) = \sqrt t has \varphi'(t) = \frac{1}{2\sqrt t}, which is undefined and unbounded at t = 0, so Theorem 7.3.8 does not apply.

Instead, let g(t) := \frac{\sin\sqrt t}{\sqrt t} on (0,4] and g(0) := 1. Since \lvert\sin u\rvert \le \lvert u\rvert, \lvert g(t)\rvert \le 1 is bounded on [0,4] and continuous on [c,4] for every c \in (0,4), so g \in \mathcal{R}[0,4] by Exercise 7.2.11 (Q28). Define F(t) := -2\cos\sqrt t on [0,4]; then F is continuous on [0,4] and F'(t) = g(t) for all t \in (0,4] (E = \{0\}). By FTC 1 (Theorem 7.3.1), \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = F(4) - F(0) = -2\cos 2 - (-2\cos 0) = 2(1 - \cos 2) \approx 2.832294.
 \int_1^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = 2(\cos 1 - \cos 2) \approx 1.912898;\quad \int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt = 2(1 - \cos 2) \approx 2.832294. Marks: statement & proof of 7.3.8 2 · substitution evaluations 2 · [0,4] explanation & FTC 1 evaluation 2(1-\cos 2) 2. </section>
<section class="page"> 31PYQ MAT-22 Q6 State Lebesgue's Integrability Criterion (7.3.12) and apply it to five benchmark functions (Examples 7.3.13(a)–(e)). Know firstT27T26T08 Theorem 7.3.12 · Lebesgue's Integrability Criterion (Bartle p. 221)A bounded function f : [a,b] \to \mathbb{R} is Riemann integrable if and only if it is continuous almost everywhere on [a,b] (equivalently, the set D(f) of points of discontinuity of f on [a,b] is a null set). SkeletonFor each function on [0,1], check (1) whether it is bounded, and (2) whether its discontinuity set D is a null set. Applications (Bartle Examples 7.3.13(a)–(e)): <ul> (a) Any step function \varphi on [0,1]: \varphi takes finitely many values, so it is bounded, and it is discontinuous at most at the finitely many endpoints of its constant intervals. Since every finite set is a null set, \varphi \in \mathcal{R}[0,1]. (b) Any monotone function f on [0,1]: f is bounded between f(0) and f(1), and by Theorem 5.6.4 (P10) its set of discontinuities is at most countable, hence a null set (T26). By Theorem 7.3.12, f \in \mathcal{R}[0,1]. (c) G(1/n) = 1/n (n \in \mathbb{N}), G(x) = 0 elsewhere on [0,1] (Example 7.1.4(d)): G is bounded (0 \le G(x) \le 1), continuous at x = 0 (since \lim_{x\to 0} G(x) = 0 = G(0)) and at every point outside \{1/n : n\in\mathbb{N}\}, and discontinuous precisely on the countable set D = \{1, \tfrac12, \tfrac13, \dots, \tfrac1n, \dots\}. Since every countable set is a null set, G \in \mathcal{R}[0,1]. (d) Dirichlet's function f on [0,1] C7: f is bounded (0 \le f \le 1), but f is discontinuous at every point of [0,1]. Since D = [0,1] is not a null set, f \notin \mathcal{R}[0,1]. (e) Thomae's function h on [0,1] C8: h is bounded (0 \le h \le 1), continuous at every irrational in [0,1], and discontinuous precisely on D = \mathbb{Q} \cap [0,1]. By Example 7.3.11, \mathbb{Q}\cap[0,1] is a null set, so h \in \mathcal{R}[0,1]. Marks: statement of Theorem 7.3.12 1 · 1 mark for each of the five cases (a)–(e).
32PYQ MAT-21 Q4 State and prove the Composition Theorem (7.3.14); counterexample when \varphi is not continuous. Know firstT28T27T26P07 Theorem 7.3.14 · Composition Theorem (Bartle p. 222)Let f \in \mathcal{R}[a,b] with f([a,b]) \subseteq [c,d], and let \varphi : [c,d] \to \mathbb{R} be continuous. Then the composition \varphi \circ f belongs to \mathcal{R}[a,b]. Proof of Theorem 7.3.14 (Bartle p. 222)
If f is continuous at a point u \in [a,b], then since \varphi is continuous at f(u) \in [c,d], the composite function \varphi \circ f is also continuous at u.<span

class="why">Composition of continuous functions is continuous (P07). Consequently, the set D of points of discontinuity of \varphi \circ f on [a,b] is contained in the set D(f) of points of discontinuity of f.Contrapositive of Step 1: u \in D \Rightarrow u \in D(f). Since f \in \mathcal{R}[a,b], Lebesgue's Integrability Criterion 7.3.12 implies that D(f) is a null set. Since any subset of a null set is a null set (T26), D is a null set.Theorem 7.3.12 (\Rightarrow) and subset property of null sets. Moreover, since \varphi is continuous on the closed bounded interval [c,d], \varphi is bounded on [c,d] (Theorem 5.3.2), so \varphi \circ f is bounded on [a,b]. By Lebesgue's Integrability Criterion 7.3.12, \varphi \circ f \in \mathcal{R}[a,b]. ∎Boundedness Theorem 5.3.2 (P07) and Theorem 7.3.12 (\Leftarrow).   Counterexample showing continuity of \varphi cannot be dropped (Exercise 7.3.22): Let h \in \mathcal{R}[0,1] be Thomae's function C8, and define \varphi : [0,1] \to \mathbb{R} by \varphi(0) := 0 and \varphi(y) := 1 for y \in (0,1]. Note that \varphi \in \mathcal{R}[0,1] (step function, discontinuous only at 0). However, for x \in [0,1], (\varphi \circ h)(x) = 1 if x \in \mathbb{Q}\cap[0,1] (where h(x) = 1/n > 0) and (\varphi \circ h)(x) = \varphi(0) = 0 if x \in [0,1]\setminus\mathbb{Q}: thus \varphi \circ h is Dirichlet's function C7, which is not in \mathcal{R}[0,1]. Marks: statement 1 · D(\varphi\circ f) \subseteq D(f) null 2 · boundedness & Lebesgue conclusion 1.5 · counterexample 1.5. </section>
<section class="page"> 33PYQ MTS-25 Q8also MTS-22 Q7, MAT-21 Q20 If f \in \mathcal{R}[a,b], then \lvert f\rvert \in \mathcal{R}[a,b] and \big\lvert\int_a^b f\big\rvert \le \int_a^b \lvert f\rvert \le M(b-a) (Corollary 7.3.15); converse false. Know firstT29T28T08T07 Corollary 7.3.15 (Bartle p. 222)Suppose f \in \mathcal{R}[a,b]. Then its absolute value \lvert f\rvert is in \mathcal{R}[a,b], and
\left\lvert\int_a^b f\right\rvert \le \int_a^b \lvert f\rvert \le M(b - a),
where \lvert f(x)\rvert \le M for all x \in [a,b]. Proof of Corollary 7.3.15 (Bartle p. 222) Since f \in \mathcal{R}[a,b], Theorem 7.1.6 implies that f is bounded on [a,b], say \lvert f(x)\rvert \le M for all x \in [a,b], so f([a,b]) \subseteq [-M,M].Boundedness Theorem 7.1.6 (T08). The function \varphi(t) := \lvert t\rvert is continuous on [-M,M]. By the Composition Theorem 7.3.14, \lvert f\rvert = \varphi \circ f belongs to \mathcal{R}[a,b].Composition Theorem 7.3.14 (T28). Since -\lvert f(x)\rvert \le f(x) \le \lvert f(x)\rvert for all x \in [a,b], Theorem 7.1.5(a) and (c) give
-\int_a^b \lvert f\rvert \le \int_a^b f \le \int_a^b \lvert f\rvert, \quad \text{and hence} \quad \left\lvert\int_a^b f\right\rvert \le \int_a^b \lvert f\rvert.
Linearity and monotonicity of the integral (7.1.5a,c, T07) and P05. Since \lvert f(x)\rvert \le M for all x \in [a,b], Theorem 7.1.5(c) and Example 7.1.4(a) give \int_a^b \lvert f\rvert \le \int_a^b M = M(b - a). ∎Constant integral (T05).   Converse is false: Define f : [0,1] \to \mathbb{R} by f(x) := 1 if x \in \mathbb{Q}\cap[0,1] and f(x) := -1 if x \in [0,1]\setminus\mathbb{Q}. Then \lvert f(x)\rvert = 1 for all x \in [0,1], so \lvert f\rvert \in \mathcal{R}[0,1], but f \notin \mathcal{R}[0,1] (rational tags give

S = 1, irrational tags give S = -1). Marks: \lvert f\rvert \in \mathcal{R}[a,b] via 7.1.6 & 7.3.14 2 · \lvert\int f\rvert \le \int\lvert f\rvert \le M(b-a) 2.5 · counterexample for converse 1.5.
34PYQ MTS-26 Q6also MAT-22 Q17, MAT-21 Q18 State and prove the Product Theorem (7.3.16); show \int_a^b fg \ne \left(\int_a^b f\right)\left(\int_a^b g\right) in general. Know firstT30T28T08T07 Theorem 7.3.16 · Product Theorem (Bartle p. 222)If f and g belong to \mathcal{R}[a,b], then the product fg belongs to \mathcal{R}[a,b]. Proof of Theorem 7.3.16 (Bartle p. 222) Since f, g \in \mathcal{R}[a,b], Theorem 7.1.5(b) implies f + g \in \mathcal{R}[a,b]. By the Boundedness Theorem 7.1.6, f, g, and f+g are bounded on [a,b]; choose M > 0 so that f([a,b]), g([a,b]), and (f+g)([a,b]) are all contained in [-M,M].Theorem 7.1.5(b) and Boundedness Theorem 7.1.6 (T08). The squaring function \varphi(t) := t^2 is continuous on [-M,M]. By the Composition Theorem 7.3.14, f^2 = \varphi \circ f, g^2 = \varphi \circ g, and (f+g)^2 = \varphi \circ (f+g) all belong to \mathcal{R}[a,b].Composition Theorem 7.3.14 (T28) with \varphi(t) = t^2. Since we have the algebraic identity
fg = \frac12\big[(f + g)^2 - f^2 - g^2\big],
Theorem 7.1.5(a) and (b) imply that fg \in \mathcal{R}[a,b]. ∎Linearity of the integral (Theorem 7.1.5, T07).
Why \int_a^b fg \ne \left(\int_a^b f\right)\left(\int_a^b g\right) in general: Let f(x) = g(x) = x on [0,1]. Then \int_0^1 fg = \int_0^1 x^2\,dx = \tfrac13, whereas \left(\int_0^1 f\right)\left(\int_0^1 g\right) = \tfrac12 \cdot \tfrac12 = \tfrac14 \ne \tfrac13. Marks: statement 1 · f^2, g^2, (f+g)^2 \in \mathcal{R}[a,b] via 7.3.14 2.5 · polarisation identity & 7.1.5 1.5 · counterexample for integral of product 1. </section>
<section class="page"> 35 State and prove Integration by Parts (Theorem 7.3.17); evaluate \int_0^1 x e^x\,dx. Know firstT31T30T20P11 Theorem 7.3.17 · Integration by Parts (Bartle p. 222, eq. (7))Let F, G be differentiable on [a,b] and let f := F' and g := G' belong to \mathcal{R}[a,b]. Then
\int_a^b fG = \Big[FG\Big]_a^b - \int_a^b Fg.
 Proof of Theorem 7.3.17 (Bartle p. 223) By the Product Rule 6.1.3(c) (P11), the product FG is differentiable on [a,b] and
(FG)'(x) = F'(x)G(x) + F(x)G'(x) = f(x)G(x) + F(x)g(x) \quad \text{for all } x \in [a,b].
Product Rule 6.1.3(c) (P11). Since F and G are differentiable on [a,b], they are continuous on [a,b] (Theorem 6.1.2) and hence belong to \mathcal{R}[a,b] (Theorem 7.2.7). Since f, g \in \mathcal{R}[a,b] by hypothesis, the Product Theorem 7.3.16 implies that fG \in \mathcal{R}[a,b] and Fg \in \mathcal{R}[a,b], so (FG)' = fG + Fg \in \mathcal{R}[a,b] by Theorem 7.1.5(b).Theorems 6.1.2, 7.2.7, 7.3.16 (T30), and 7.1.5(b). Applying the First Form of the Fundamental Theorem 7.3.1 (with E = \varnothing) to FG gives
\Big[FG\Big]_a^b = \int_a^b (FG)' = \int_a^b fG + \int_a^b Fg, \quad \text{whence} \quad \int_a^b

fG = \Big[FG\Big]_a^b - \int_a^b Fg. \quad \blacksquare
Theorem 7.3.1 (T20) and Theorem 7.1.5(b).
Evaluation of \int_0^1 x e^x\,dx: Put F(x) := e^x (f(x) = F'(x) = e^x \in \mathcal{R}[0,1]) and G(x) := x (g(x) = G'(x) = 1 \in \mathcal{R}[0,1]). By Theorem 7.3.17, \int_0^1 e^x\cdot x\,dx = \big[x e^x\big]_0^1 - \int_0^1 e^x\cdot 1\,dx = (1\cdot e - 0) - \big[e^x\big]_0^1 = e - (e - 1) = 1.
 \displaystyle\int_0^1 x e^x\,dx = 1. Marks: statement 1 · Product Rule + Product Theorem step 2 · FTC 1 conclusion 1.5 · \int_0^1 x e^x\,dx = 1 1.5.
36PYQ MTS-25 Q19also MAT-22 Q32 State and prove Taylor's Theorem with the Integral Remainder (7.3.18); verify for f(x) = e^x on [0,1] with n = 1, 2. Know firstT32T31T20 Theorem 7.3.18 · Taylor's Theorem with the Remainder (Bartle p. 223, eqs. (8)–(9))Suppose that f', \dots, f^{(n)}, f^{(n+1)} exist on [a,b] and that f^{(n+1)} \in \mathcal{R}[a,b]. Then we have
f(b) = f(a) + \frac{f'(a)}{1!}(b - a) + \cdots + \frac{f^{(n)}(a)}{n!}(b - a)^n + R_n,
where the remainder is given by
R_n = \frac{1}{n!}\int_a^b f^{(n+1)}(t)\,(b - t)^n\,dt.
 Proof of Theorem 7.3.18 (Bartle p. 223, with misprint corrected) Apply Integration by Parts (Theorem 7.3.17) to R_n with F(t) := f^{(n)}(t) (so f(t) = F'(t) = f^{(n+1)}(t) \in \mathcal{R}[a,b]) and G(t) := (b - t)^n/n! (so g(t) = G'(t) = -(b - t)^{n-1}/(n - 1)! \in \mathcal{R}[a,b]).Check hypotheses of Theorem 7.3.17 (T31). Since G(b) = 0 and G(a) = (b - a)^n/n!, Theorem 7.3.17 gives
R_n = \left[\frac{f^{(n)}(t)(b-t)^n}{n!}\right]_{t=a}^{t=b} - \int_a^b f^{(n)}(t)\left(-\frac{(b-t)^{n-1}}{(n-1)!}\right)dt = -\frac{f^{(n)}(a)}{n!}(b - a)^n + R_{n-1},
where R_{n-1} = \dfrac{1}{(n-1)!}\int_a^b f^{(n)}(t)\,(b - t)^{n-1}\,dt.Boundary evaluation at t=b and t=a. (Note: Bartle p. 223 prints (b-a)^{n-1} in the first displayed integral; the correct integrand is (b-t)^{n-1}.) For each k \le n, f^{(k)} is differentiable on [a,b], hence continuous (Theorem 6.1.2) and in \mathcal{R}[a,b] (Theorem 7.2.7). Repeating Step 2 n times yields R_{k-1} = \frac{f^{(k)}(a)}{k!}(b-a)^k + R_k down to R_0 = \int_a^b f'(t)\,dt = f(b) - f(a) (by FTC 1, Theorem 7.3.1), which rearranges to Taylor's formula. ∎Repeated Integration by Parts and FTC 1 (T20).   Verification for f(x) = e^x, a = 0, b = 1: Here f^{(k)}(0) = 1 for all k. For n = 1: T_1 = 1 + 1 = 2 and R_1 = \int_0^1 e^t(1-t)\,dt = \big[e^t(1-t)\big]_0^1 + \int_0^1 e^t\,dt = -1 + (e - 1) = e - 2 \approx 0.718282, so T_1 + R_1 = e. For n = 2: T_2 = 1 + 1 + \tfrac12 = \tfrac52 and R_2 = -\tfrac12 + R_1 = e - \tfrac52 \approx 0.218282, so T_2 + R_2 = e. R_1 = e - 2 \approx 0.718282;\quad R_2 = e - \frac52 \approx 0.218282. Marks: statement 1.5 · IBP recurrence R_n = -\frac{f^{(n)}(a)}{n!}(b-a)^n + R_{n-1} 2 · induction to R_0 = f(b)-f(a) 1 · e^x check 1.5.
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<!-- SECTION C (Q37-Q40) --> <section class="page"> Section C · Essay answers (Q37–Q40)<small>10 marks each. Answer any two.</small>
37PYQ MTS-25 Q18also MAT-21 Q30, MTS-22 Q19, MAT-22 MCQ 21, MAT-21 MCQ 15 Continuous on [a,b] \Rightarrow Riemann integrable

(Theorem 7.2.7); where hypotheses are used; converses. Know firstT15T12T14P07P08 Theorem 7.2.7 · Integrability of Continuous Functions (Bartle p. 212)If f : [a,b] \to \mathbb{R} is continuous on [a,b], then f \in \mathcal{R}[a,b]. Skeleton1. Use UCT 5.4.3 to get \delta_\varepsilon > 0 so \lvert u - v\rvert < \delta_\varepsilon \Rightarrow \lvert f(u) - f(v)\rvert < \varepsilon/(b-a). 2. Pick a partition P with \lVert P\rVert < \delta_\varepsilon. 3. Use Max–Min 5.3.4 on each I_i to get u_i, v_i \in I_i with f(u_i) = \min_{I_i} f, f(v_i) = \max_{I_i} f. 4. Define step functions \alpha_\varepsilon, \omega_\varepsilon with heights f(u_i), f(v_i) and apply the Squeeze Theorem 7.2.3. Proof of Theorem 7.2.7 (Bartle p. 212) Since f is continuous on the closed bounded interval [a,b], the Uniform Continuity Theorem 5.4.3 (P08) implies that f is uniformly continuous on [a,b]. Therefore, given any \varepsilon > 0, there exists \delta_\varepsilon > 0 such that if u, v \in [a,b] and \lvert u - v\rvert < \delta_\varepsilon, then
\lvert f(u) - f(v)\rvert < \frac{\varepsilon}{b - a}.
Uniform Continuity Theorem 5.4.3 applied with tolerance \varepsilon/(b-a) > 0. Let P = \{I_i\}_{i=1}^n be a partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, where I_i = [x_{i-1},x_i].Choose an equal partition with n > (b-a)/\delta_\varepsilon. Since f is continuous on each closed bounded subinterval I_i = [x_{i-1},x_i], the Maximum–Minimum Theorem 5.3.4 (P07) implies that there exist points u_i, v_i \in I_i where f attains its minimum and maximum values on I_i, respectively:
f(u_i) \le f(x) \le f(v_i) \quad \text{for all } x \in I_i.
Maximum–Minimum Theorem 5.3.4 on each I_i. Define step functions \alpha_\varepsilon, \omega_\varepsilon : [a,b] \to \mathbb{R} by
\alpha_\varepsilon(x) := f(u_i) \text{ for } x \in [x_{i-1},x_i)\ (i < n), \quad \alpha_\varepsilon(x) := f(u_n) \text{ for } x \in [x_{n-1},x_n], \omega_\varepsilon(x) := f(v_i) \text{ for } x \in [x_{i-1},x_i)\ (i < n), \quad \omega_\varepsilon(x) := f(v_n) \text{ for } x \in [x_{n-1},x_n].
By Theorem 7.2.5, \alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b], and by Step 3, \alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) for all x \in [a,b].Step-function integrability (Theorem 7.2.5, T14). Since u_i, v_i \in I_i and \lVert P\rVert < \delta_\varepsilon, we have \lvert u_i - v_i\rvert \le x_i - x_{i-1} < \delta_\varepsilon, so f(v_i) - f(u_i) < \dfrac{\varepsilon}{b-a} by Step 1. Therefore, by formula (5) of Theorem 7.2.5,
0 \le \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) = \sum_{i=1}^n \big(f(v_i) - f(u_i)\big)(x_i - x_{i-1}) < \sum_{i=1}^n \frac{\varepsilon}{b - a}(x_i - x_{i-1}) = \frac{\varepsilon}{b - a}(b - a) = \varepsilon.
Theorem 7.2.5 formula and telescoping \sum(x_i - x_{i-1}) = b - a (M01). Since \varepsilon > 0 is arbitrary, the Squeeze Theorem 7.2.3 implies that f \in \mathcal{R}[a,b]. ∎Squeeze Theorem 7.2.3 (T12).   Where each hypothesis was used: Continuity of f and the fact that [a,b] is closed and bounded are used twice: (i) in Step 1 to invoke the Uniform Continuity Theorem 5.4.3 so a single \delta_\varepsilon works across all subintervals, and (ii) in Step 3 to invoke the Maximum–Minimum Theorem 5.3.4 on each closed subinterval I_i. Converses are FALSE (MAT-22 MCQ 21, MAT-21

MCQ 15): Option (A) "f continuous on [a,b] \Rightarrow f \in \mathcal{R}[a,b]" is true, while (B) and (C) are false: the step function g(x) = 2 on [0,1], 3 on (1,3] (or Thomae's function h C8) belongs to \mathcal{R}, yet is discontinuous and hence not uniformly continuous. Marks: UCT step 2 · Max–Min step 2 · step functions \alpha_\varepsilon, \omega_\varepsilon 2 · \int(\omega_\varepsilon-\alpha_\varepsilon) < \varepsilon & Squeeze 2 · where hypotheses used & converses 2. </section>
<section class="page"> 38PYQ MTS-26 Q19also MAT-22 Q30, MAT-22 MCQ 22 Monotone on [a,b] \Rightarrow Riemann integrable (Theorem 7.2.8); staircase calculation for f(x) = x^2 on [0,1] (n=4); converse. Know firstT16T12T14M05 Theorem 7.2.8 · Integrability of Monotone Functions (Bartle p. 212)If f : [a,b] \to \mathbb{R} is monotone on [a,b], then f \in \mathcal{R}[a,b]. SkeletonAssume f is increasing on [a,b]. Partition [a,b] into n equal subintervals of length (b-a)/n. Define step functions \alpha(x) = f(x_{k-1}) and \omega(x) = f(x_k) on I_k, telescope \int_a^b(\omega-\alpha) = \frac{b-a}{n}(f(b)-f(a)), choose n > (b-a)(f(b)-f(a))/\varepsilon, and apply the Squeeze Theorem 7.2.3. Proof of Theorem 7.2.8 (Bartle p. 212) Assume f is increasing on I = [a,b] (if f is decreasing, then -f is increasing, so -f \in \mathcal{R}[a,b] and hence f = -(-f) \in \mathcal{R}[a,b] by Theorem 7.1.5(a)). For n \in \mathbb{N}, partition [a,b] into n equal subintervals I_k = [x_{k-1},x_k] with x_k - x_{k-1} = \dfrac{b-a}{n} (k = 1, \dots, n).Equal subinterval lengths (b-a)/n allow the step integrals to telescope. Since f is increasing on I_k = [x_{k-1},x_k], its minimum on I_k is f(x_{k-1}) and its maximum on I_k is f(x_k):
f(x_{k-1}) \le f(x) \le f(x_k) \quad \text{for all } x \in I_k.
Definition of an increasing function (P10). Define step functions \alpha, \omega : [a,b] \to \mathbb{R} by \alpha(x) := f(x_{k-1}) and \omega(x) := f(x_k) for x \in [x_{k-1},x_k) (k = 1,\dots,n-1), and \alpha(x) := f(x_{n-1}), \omega(x) := f(x_n) for x \in [x_{n-1},x_n]. Then \alpha, \omega \in \mathcal{R}[a,b] by Theorem 7.2.5 and \alpha(x) \le f(x) \le \omega(x) for all x \in [a,b].Theorem 7.2.5 (T14). By Theorem 7.2.5,
\int_a^b \alpha = \frac{b - a}{n}\big(f(x_0) + f(x_1) + \cdots + f(x_{n-1})\big), \qquad \int_a^b \omega = \frac{b - a}{n}\big(f(x_1) + \cdots + f(x_{n-1}) + f(x_n)\big).
Subtracting and cancelling the interior terms f(x_1), \dots, f(x_{n-1}) yields
\int_a^b (\omega - \alpha) = \frac{b - a}{n}\big(f(x_n) - f(x_0)\big) = \frac{b - a}{n}\big(f(b) - f(a)\big).
Telescoping sum (M01, M05) with x_0 = a and x_n = b. Given \varepsilon > 0, choose n \in \mathbb{N} with n > \dfrac{(b - a)(f(b) - f(a))}{\varepsilon}. Then \int_a^b (\omega - \alpha) < \varepsilon, so by the Squeeze Theorem 7.2.3, f \in \mathcal{R}[a,b]. ∎Archimedean property (P13) and Squeeze Theorem 7.2.3 (T12).   Staircase illustration for f(x) = x^2 on [0,1] with n = 4: Here P_4 = (0, \tfrac14, \tfrac12, \tfrac34, 1) has width \Delta x_k = \tfrac14, and f(x) = x^2 takes values 0, \tfrac1{16}, \tfrac4{16}, \tfrac9{16}, 1 at the partition points:
\int_0^1 \alpha = \frac14\left(0 + \frac1{16} + \frac4{16} + \frac9{16}\right) = \frac{14}{64} =

\frac{7}{32} = 0.21875, \qquad \int_0^1 \omega = \frac14\left(\frac1{16} + \frac4{16} + \frac9{16} + 1\right) = \frac{30}{64} = \frac{15}{32} = 0.46875,
 and \int_0^1(\omega - \alpha) = 0.46875 - 0.21875 = 0.25 = \frac{1-0}{4}(1 - 0). Converse is FALSE (MAT-22 MCQ 22): Option (A) is true, while (B) "f \in \mathcal{R}[a,b] \Rightarrow f is monotone" is false: f(x) = \sin x on [0,\pi] (or x^2 on [-1,1], or Thomae's function C8) is in \mathcal{R} but not monotone. For x^2 on [0,1] (n=4): \int_0^1 \alpha = 0.21875, \int_0^1 \omega = 0.46875, difference = 0.25. Marks: equal partition & \alpha,\omega setup 2.5 · telescoping \int(\omega-\alpha) = \frac{b-a}{n}(f(b)-f(a)) 2.5 · choice of n & Squeeze 2 · x^2 (n=4) calculation 2 · converse false 1. </section>
<section class="page"> 39PYQ MTS-26 Q25also MAT-22 Q37a, MTS-22 Q25 State and prove the Additivity Theorem (Theorem 7.2.9). Know firstT17T10T08T03M08 Theorem 7.2.9 · Additivity Theorem (Bartle p. 213)Let f : [a,b] \to \mathbb{R} and let c \in (a,b). Then f \in \mathcal{R}[a,b] if and only if its restrictions to [a,c] and to [c,b] are both Riemann integrable. In this case
\int_a^b f = \int_a^c f + \int_c^b f.
 Skeleton(\Leftarrow) Take \delta', \delta'' for f_1, f_2 with \varepsilon/3; state f is bounded by M > 0 (7.1.6 on each piece); put \delta_\varepsilon = \min\{\delta',\delta'',\varepsilon/(6M)\}. Split into Case (i) c \in Q and Case (ii) c \in (x_{k-1},x_k). (\Rightarrow) Use Cauchy 7.2.1 by extending \dot P_1, \dot Q_1 on [a,c] with the same \dot R on [c,b]. Proof of Theorem 7.2.9 (Bartle pp. 213–214) (\Leftarrow) Suppose f_1 := f|_{[a,c]} \in \mathcal{R}[a,c] with integral L_1 and f_2 := f|_{[c,b]} \in \mathcal{R}[c,b] with integral L_2. Given \varepsilon > 0, choose \delta' > 0 such that \lVert P_1\rVert < \delta' \Rightarrow \lvert S(f_1;\dot P_1) - L_1\rvert < \varepsilon/3, and choose \delta'' > 0 such that \lVert P_2\rVert < \delta'' \Rightarrow \lvert S(f_2;\dot P_2) - L_2\rvert < \varepsilon/3.Definition 7.1.1 applied on [a,c] and [c,b] with tolerance \varepsilon/3. Since f_1 \in \mathcal{R}[a,c] and f_2 \in \mathcal{R}[c,b], Theorem 7.1.6 implies that f is bounded on [a,c] and on [c,b], and hence bounded on [a,b]. Choose M > 0 such that \lvert f(x)\rvert \le M for all x \in [a,b] (if f \equiv 0, take any M > 0), and define
\delta_\varepsilon := \min\left\{\delta',\, \delta'',\, \frac{\varepsilon}{6M}\right\} > 0.
Let \dot Q = \{(I_i,t_i)\}_{i=1}^m be any tagged partition of [a,b] with \lVert Q\rVert < \delta_\varepsilon.Boundedness Theorem 7.1.6 (T08) and three-way minimum (M08). Case (i): c is a partition point of \dot Q. Split \dot Q at c into a tagged partition \dot Q_1 of [a,c] and a tagged partition \dot Q_2 of [c,b]. Since S(f;\dot Q) = S(f_1;\dot Q_1) + S(f_2;\dot Q_2) and \lVert Q_1\rVert < \delta', \lVert Q_2\rVert < \delta'', Step 1 and the Triangle Inequality give
\lvert S(f;\dot Q) - (L_1 + L_2)\rvert \le \lvert S(f_1;\dot Q_1) - L_1\rvert + \lvert S(f_2;\dot Q_2) - L_2\rvert < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon.
Direct split when c is a cut point. Case (ii): c is not a partition point of \dot Q. Then c \in (x_{k-1},x_k) for some k \le m. Define tagged partitions \dot Q_1 of [a,c] and \dot Q_2 of [c,b] by
\dot Q_1 := \{(I_1,t_1),\dots,(I_{k-1},t_{k-1}),([x_{k-1},c],c)\}, \qquad \dot Q_2 :=

\{([c,x_k],c),(l_{k+1},t_{k+1}),\dots,(l_m,t_m)\}.
Then all terms of S(f;\dot Q) and S(f_1;\dot Q_1) + S(f_2;\dot Q_2) cancel except on l_k, giving S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2) = \big(f(t_k) - f(c)\big)(x_k - x_{k-1}), so \lvert S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2)\rvert \le 2M(x_k - x_{k-1}) \le 2M\lVert Q\rVert < 2M\cdot\dfrac{\varepsilon}{6M} = \dfrac{\varepsilon}{3}.Only interval l_k is modified; \lvert f(t_k) - f(c)\rvert \le 2M and x_k - x_{k-1} < \varepsilon/(6M). Since \lVert Q_1\rVert \le \lVert Q\rVert < \delta' and \lVert Q_2\rVert \le \lVert Q\rVert < \delta", Step 1 gives \lvert S(f_1;\dot Q_1) - L_1\rvert < \varepsilon/3 and \lvert S(f_2;\dot Q_2) - L_2\rvert < \varepsilon/3. By the Triangle Inequality,
\lvert S(f;\dot Q) - (L_1 + L_2)\rvert \le \lvert S(f;\dot Q) - S(f_1;\dot Q_1) - S(f_2;\dot Q_2)\rvert + \lvert S(f_1;\dot Q_1) - L_1\rvert + \lvert S(f_2;\dot Q_2) - L_2\rvert < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.
Since \varepsilon > 0 is arbitrary, f \in \mathcal{R}[a,b] and \int_a^b f = L_1 + L_2 = \int_a^c f + \int_c^b f.Triangle Inequality and Definition 7.1.1. (\Rightarrow) Suppose f \in \mathcal{R}[a,b] and let \varepsilon > 0. By the Cauchy Criterion 7.2.1, there exists \eta_\varepsilon > 0 such that any tagged partitions \dot P, \dot Q of [a,b] with norms <\eta_\varepsilon satisfy \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon.Cauchy Criterion 7.2.1 (\Rightarrow). Let \dot P_1, \dot Q_1 be any tagged partitions of [a,c] with \lVert P_1\rVert < \eta_\varepsilon and \lVert Q_1\rVert < \eta_\varepsilon. Adjoin the same tagged partition \dot R of [c,b] with \lVert R\rVert < \eta_\varepsilon to both \dot P_1 and \dot Q_1 to obtain tagged partitions \dot P = \dot P_1 \cup \dot R and \dot Q = \dot Q_1 \cup \dot R of [a,b] with \lVert P\rVert, \lVert Q\rVert < \eta_\varepsilon. Then
\lvert S(f_1;\dot P_1) - S(f_1;\dot Q_1)\rvert = \lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon.
By the Cauchy Criterion 7.2.1, f_1 \in \mathcal{R}[a,c]. By the same argument, f_2 \in \mathcal{R}[c,b], and \int_a^b f = \int_a^c f + \int_c^b f follows from the (\Leftarrow) part. ∎The [c,b] terms S(f;\dot R) cancel identically; Cauchy Criterion 7.2.1 (\Leftarrow).   Marks: statement 1 · boundedness M>0 & \delta_\varepsilon = \min\{\delta',\delta",\varepsilon/(6M)\} 2 · Case (i) & Case (ii) estimates 4 · (\Rightarrow) Cauchy extension 3. </section>
<section class="page"> 40PYQ MTS-25 Q25also MAT-21 Q37a, MAT-22 Q33 State and prove the Second Form of the Fundamental Theorem of Calculus (Theorems 7.3.5 & 7.3.6); counterexample at a discontinuity. Know firstT23T24T22T17T07 Theorem 7.3.5 & Theorem 7.3.6 · FTC Second Form (Bartle p. 219) Theorem 7.3.5: Let f \in \mathcal{R}[a,b] and let f be continuous at a point c \in [a,b]. Then the indefinite integral F(z) := \int_a^z f (z \in [a,b]) is differentiable at c and F'(c) = f(c). Theorem 7.3.6: If f is continuous on [a,b], then the indefinite integral F(z) := \int_a^z f is differentiable on [a,b] and F'(x) = f(x) for all x \in [a,b]. SkeletonFor c \in [a,b), given \varepsilon > 0 use continuity of f at c to find \eta_\varepsilon > 0 so c \le x < c + \eta_\varepsilon \Rightarrow f(c) - \varepsilon < f(x) < f(c) + \varepsilon. For 0 < h < \eta_\varepsilon, write F(c+h) - F(c) = \int_c^{c+h} f, sandwich between (f(c)-\varepsilon)h and (f(c)+\varepsilon)h, divide by h, and repeat for h <

0. Proof of Theorems 7.3.5 and 7.3.6 (Bartle pp. 219–220) Suppose first that c \in [a,b) and consider the right-hand derivative of F at c. Since f is continuous at c, given \varepsilon > 0 there exists \eta_\varepsilon > 0 such that if c \le x < c + \eta_\varepsilon (x \in [a,b]), then f(c) - \varepsilon < f(x) < f(c) + \varepsilon.
Definition 5.1.1 of continuity of f at c. Let h satisfy 0 < h < \eta_\varepsilon (with c + h \le b). By the Additivity Theorem 7.2.9 (and Corollary 7.2.10), f is integrable on [a,c], [a,c+h], and [c,c+h], and
F(c + h) - F(c) = \int_a^{c+h} f - \int_a^c f = \int_c^{c+h} f.
Additivity Theorem 7.2.9 (T17). On the interval [c, c+h], the inequality of Step 1 holds, so Theorem 7.1.5(c) and Example 7.1.4(a) yield (f(c) - \varepsilon)\cdot h \le F(c + h) - F(c) = \int_c^{c+h} f \le (f(c) + \varepsilon)\cdot h.
Monotonicity of the integral (Theorem 7.1.5(c), T07) and constant integral \int_c^{c+h} k = kh (T05). Dividing this inequality by h > 0 and subtracting f(c) gives
\left\lvert\frac{F(c + h) - F(c)}{h} - f(c)\right\rvert \le \varepsilon.
Since \varepsilon > 0 is arbitrary, the right-hand limit is \lim_{h\to 0^+} \dfrac{F(c+h) - F(c)}{h} = f(c).Definition of right-hand derivative. If c \in (a,b], the left-hand limit is proved identically: for -\eta_\varepsilon < h < 0 (where f(c)-\varepsilon < f(x) < f(c)+\varepsilon on (c-\eta_\varepsilon,c]), on [c+h,c] of length -h > 0 we have (f(c)-\varepsilon)(-h) \le \int_{c+h}^c f = F(c) - F(c+h) \le (f(c)+\varepsilon)(-h); dividing by -h > 0 gives \left\lvert\frac{F(c+h)-F(c)}{h} - f(c)\right\rvert \le \varepsilon, so \lim_{h\to 0^-} \frac{F(c+h)-F(c)}{h} = f(c). Hence F is differentiable at c and F'(c) = f(c).Agreement of left and right derivatives at interior points (and one-sided derivative at a or b). Proof of Theorem 7.3.6: If f is continuous at every point of [a,b], then f \in \mathcal{R}[a,b] by Theorem 7.2.7, and applying Theorem 7.3.5 at each x \in [a,b] shows that F is differentiable on [a,b] with F'(x) = f(x) for all x \in [a,b]. ∎Pointwise application of Theorem 7.3.5 across [a,b]. (Note: MAT-22 Q33 states Theorem 7.3.5 in full.)   Example where F fails to be differentiable at 0 (Example 7.3.7(a), C20): Let f(x) := \operatorname{sgn} x on [-1,1]. Then f \in \mathcal{R}[-1,1] (step function), and its indefinite integral with basepoint -1 is F(x) = \int_{-1}^x \operatorname{sgn} = \lvert x\rvert - 1. At c = 0 (where f is discontinuous), F'_-(0) = -1 \ne +1 = F'_+(0), so F is not differentiable at 0. Marks: statements of 7.3.5 & 7.3.6 2 · continuity of f at c setup 1.5 · Additivity F(c+h)-F(c) = \int_c^{c+h} f 1.5 · sandwiching & division by h 2 · left-hand limit & 7.3.6 1.5 · \operatorname{sgn} counterexample 1.5. </section>
=== src/m3/figs_m3.py === """Module 3 figures: The Riemann integral (Bartle & Sherbert 4e, §7.1–§7.3). Follows brief §8 (F01–F18) and the colour/helper conventions of figs.py and figs_m2.py. Each function returns an SVG string (placed inside <figure>{{fig:NAME}}<figcaption>...</figcaption></figure>)."""
import math
from figs import *
def F01_partition_norm():
    p = Plot(-0.2, 4.3, -0.9, 0.8, 360, 145, (18, 12, 14, 18))
    p.seg(0, 0, 4, 0, INK, 2.2)
    pts = [0, 1, 1.5, 2, 3.4, 4]
    for i, x in enumerate(pts):
        p.line(p.X(x), p.Y(0) - 5, p.X(x), p.Y(0) + 5, INK, 1.6)
        p.dot(p.X(x), p.Y(0), BLUE, 3.0)
        p.lab(x, 0.35, f'x{sub(i)}={x:g}', INK, 9.5)
    gaps = [(0, 1, '1', GREEN), (1, 1.5, '.5', GREEN), (1.5, 2, '.5', GREEN), (2, 3.4, '‖P‖ = 1.4', ORANGE), (3.4, 4, '.6', GREEN)]
    for a, b, lab, col in gaps:
        p.brace_x(a, b, -0.32, lab, col, dy=13, size=9.5)
    p.vband(2, 3.4, ORANGE, 0.16, -0.12, 0.18)
    p.lab(2.7, 0.62, f'I{sub(4)} = [2, 3.4]', ORANGE, 10)
    return p.svg()
def F02_tagged_riemann_sum():
    p = Plot(-0.08, 1.08, -0.15, 1.08, 360, 200, (28, 12, 14, 24))
    f =

```
lambda x: x * x p.axes() pts = [0, 0.2, 0.5, 0.7, 1.0] tags = [0.1, 0.5, 0.6, 0.9] for (a, b), t in zip(zip(pts, pts[1:]), tags): p.box(a, b, f(t), AMBER, 0.34, AMBER) p.seg(t, 0, t, f(t), PURPLE, 1.1, '3 2') p.dot(p.X(t), p.Y(0), PURPLE, 2.6) p.dot(p.X(t), p.Y(f(t)), PURPLE, 3.0) p.curve(f, 0, 1, BLUE, 2.4) for x in pts: p.xtick(x, f'{x:g}', size=9.5) p.ytick(1, '1', size=9.5) p.lab(0.05, 0.92, 'f(x) = x²', BLUE, 10.5, 'start') p.lab(0.05, 0.74, 'tags tᵢ (purple dots)', PURPLE, 10, 'start') p.lab(0.05, 0.56, 'S(f; Ṗ) = 0.392 (∫ = 1/3)', AMBER, 10, 'start') return p.svg()
def _tag_panel(tags, title, s_val): p = Plot(-0.06, 1.06, -0.18, 1.08, 176, 125, (20, 10, 8, 18)) f = lambda x: x * x p.axes(xlab='', ylab='') pts = [0, 0.25, 0.5, 0.75, 1.0] for (a, b), t in zip(zip(pts, pts[1:]), tags): p.box(a, b, f(t), AMBER, 0.34, AMBER) p.dot(p.X(t), p.Y(f(t)), PURPLE, 2.5) p.hline(1 / 3, INK, 1.0, '3 3', 0, 1) p.curve(f, 0, 1, BLUE, 2.0) p.lab(0.04, 0.92, title, INK, 9.5, 'start') p.lab(0.04, 0.70, f'S = {s_val}', ORANGE, 9.5, 'start') return p
def F03_four_tag_rules(): p1 = _tag_panel([0, 0.25, 0.5, 0.75], 'left tags', '0.2188') p2 = _tag_panel([0.25, 0.5, 0.75, 1.0], 'right tags', '0.4688') p3 = _tag_panel([0.125, 0.375, 0.625, 0.875], 'midpoints', '0.3281') p4 = _tag_panel([0.2, 0.3, 0.7, 0.8], 'random tags', '0.3150') return stack([(p1, 0, 0), (p2, 182, 0), (p3, 0, 130), (p4, 182, 130)], 360, 260)
def F04_mesh_vs_count(): p = Plot(-0.06, 1.06, -1.1, 1.1, 360, 160, (18, 12, 14, 16)) p.seg(0, 0.45, 1, 0.45, INK, 2.0) for x in [0, 0.25, 0.5, 0.75, 1.0]: p.line(p.X(x), p.Y(0.45) - 4, p.X(x), p.Y(0.45) + 4, BLUE, 1.6) p.dot(p.X(x), p.Y(0.45), BLUE, 2.8) p.brace_x(0.25, 0.5, 0.15, f'‖P{sub("a")}‖ = 0.25', GREEN, dy=12, size=9.5) p.lab(0, 0.78, f'P{sub("a")}: n = 4 equal pieces', BLUE, 10, 'start')
p.vband(0.3, 1.0, PINK, 0.20, -0.68, -0.32) p.seg(0, -0.5, 1, -0.5, INK, 2.0) for x in [0, 0.1, 0.2, 0.3, 1.0]: p.line(p.X(x), p.Y(-0.5) - 4, p.X(x), p.Y(-0.5) + 4, PINK, 1.6) p.dot(p.X(x), p.Y(-0.5), PINK, 2.8) p.brace_x(0.3, 1.0, -0.82, f'‖P{sub("b")}‖ = 0.70', ORANGE, dy=12, size=9.5) p.lab(0, -0.18, f'P{sub("b")}: n = 4 pieces, wide gap [0.3, 1]', PINK, 10, 'start') return p.svg()
def F05_squeeze_step(): p = Plot(-0.08, 1.08, -0.15, 1.12, 360, 205, (28, 12, 14, 24)) p.axes() for k in range(1, 5): a, b = (k - 1) / 4, k / 4 lo, hi = (k - 1) / 4, k / 4 p.box(a, b, lo, SKY, 0.28, BLUE) p.poly([(a, lo), (b, lo), (b, hi), (a, hi)], ORANGE, 0.32, stroke=ORANGE, w=1.2) p.seg(a, lo, b, lo, BLUE, 2.2) p.seg(a, hi, b, hi, ORANGE, 2.2) p.curve(lambda x: x, 0, 1, INK, 2.2) for x in [0, 0.25, 0.5, 0.75, 1.0]: p.xtick(x, f'{x:g}', size=9.5) p.ytick(1, '1', size=9.5) p.lab(0.04, 0.96, f'ω{sub(4)} (upper, ∫ = 5/8)', ORANGE, 10, 'start') p.lab(0.04, 0.78, f'α{sub(4)} (lower, ∫ = 3/8)', BLUE, 10, 'start') p.lab(0.04, 0.60, '4 gap squares: ∫(ω−α) = 1/4 = 1/n', INK, 10, 'start') return p.svg()
def F06_cont_min_max_steps(): p = Plot(-0.2, 3.35, -0.18, 1.18, 360, 200, (26, 12, 14, 24)) p.axes() s2 = math.sqrt(0.5) pieces = [(0, math.pi / 4, 0, s2, 0, math.pi / 4), (math.pi / 4, math.pi / 2, s2, 1.0, math.pi / 4, math.pi / 2), (math.pi / 2, 3 * math.pi / 4, s2, 1.0, 3 * math.pi / 4, math.pi / 2), (3 * math.pi / 4, math.pi, 0, s2, math.pi, 3 * math.pi / 4)] for a, b, lo, hi, u, v in pieces: p.box(a, b, lo, SKY, 0.25, BLUE) p.poly([(a, lo), (b, lo), (b, hi), (a, hi)], ORANGE, 0.26, stroke=ORANGE, w=1.1) p.seg(a, lo, b, lo, BLUE, 2.0) p.seg(a, hi, b, hi, ORANGE, 2.0) p.dot(p.X(u), p.Y(lo), BLUE, 2.8) p.dot(p.X(v), p.Y(hi), ORANGE, 2.8) p.curve(math.sin, 0, math.pi, INK, 2.2) p.xtick(math.pi / 2, 'π/2', size=9.5) p.xtick(math.pi, 'π', size=9.5) p.ytick(1, '1', size=9.5) p.lab(0.08, 1.06, 'ω_ε = f(vᵢ) (max on Iᵢ)', ORANGE, 9.5, 'start') p.lab(0.08, 0.88, 'α_ε = f(uᵢ) (min on Iᵢ)', BLUE, 9.5, 'start') p.lab(2.15, 1.06, 'f(x) = sin x', INK, 10, 'start') return p.svg()
def F07_monotone_staircase(): p1 = Plot(-0.08, 1.08, -0.16, 1.12, 230, 195, (24, 12, 10, 22)) f = lambda x: x * x p1.axes() for k in range(1, 5): a, b = (k - 1) / 4, k / 4 lo, hi = f(a), f(b) p1.box(a, b, lo, SKY, 0.26, BLUE) p1.poly([(a, lo), (b, lo), (b, hi), (a, hi)], ORANGE, 0.34, stroke=ORANGE, w=1.2) p1.curve(f, 0, 1, BLUE, 2.2) p1.xtick(1, '1', size=9.5) p1.ytick(1, '1', size=9.5) p1.lab(0.05, 0.95, 'f(x) = x² (n = 4)', BLUE, 9.5, 'start') p1.lab(0.05, 0.76, '4 gap blocks ω − α', ORANGE, 9.5, 'start')
p2 = Plot(-0.1, 0.6, -0.16, 1.12, 125, 195, (18, 12, 10, 22)) p2.axes(xlab='', ylab='') for k in
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range(1, 5): lo, hi = f((k - 1) / 4), f(k / 4) p2.poly([(0.1, lo), (0.35, lo), (0.35, hi), (0.1, hi)], ORANGE, 0.38, stroke=ORANGE, w=1.2) p2.brace_x(0.1, 0.35, -0.08, '1/4', GREEN, dy=11, size=9) p2.brace_y(0, 1, 0.38, 'f(1)−f(0)=1', ORANGE, dx=4, size=8.5) p2.lab(0.22, 1.05, 'stacked = 1/4', INK, 9.5) return stack([(p1, 0, 0), (p2, 232, 0)], 360, 195)
def F08_dirichlet_tags(): pts = [i / 6 for i in range(7)] p1 = Plot(-0.06, 1.06, -0.22, 1.25, 360, 115, (26, 8, 14, 18)) p1.axes(xlab='', ylab='y') for a, b in zip(pts, pts[1:]): p1.box(a, b, 1.0, AMBER, 0.32, AMBER) t = (a + b) / 2 p1.dot(p1.X(t), p1.Y(1.0), BLUE, 2.8) p1.hline(1.0, BLUE, 1.6, '2 3', 0, 1) p1.hline(0.0, PINK, 1.6, '2 3', 0, 1) p1.ytick(1, '1', size=9.5) p1.lab(0.5, 1.10, 'Rational tags tᵢ ∈ 𝕼: every f(tᵢ) = 1 ⇒ S(f; Ṗ) = 1', BLUE, 9.8)
p2 = Plot(-0.06, 1.06, -0.22, 1.25, 360, 115, (26, 8, 14, 18)) p2.axes(xlab='x', ylab='y') for a, b in zip(pts, pts[1:]): t = a + 0.52 * (b - a) p2.dot(p2.X(t), p2.Y(0.0), PINK, 2.8) p2.hline(1.0, BLUE, 1.6, '2 3', 0, 1) p2.hline(0.0, PINK, 1.6, '2 3', 0, 1) p2.ytick(1, '1', size=9.5) p2.xtick(1, '1', size=9.5) p2.lab(0.5, 0.55, 'Irrational tags sᵢ ∉ 𝕼: every f(sᵢ) = 0 ⇒ S(f; Q̇) = 0', PINK, 9.8) return stack([(p1, 0, 0), (p2, 0, 115)], 360, 230)
def F09_thomae_popcorn(): p = Plot(-0.06, 1.06, -0.14, 1.12, 360, 200, (28, 12, 14, 24)) p.axes(ylab='h') p.hband(0, 0.2, GREEN, 0.16, 0, 1) p.hline(0.2, ORANGE, 1.3, '4 3', 0, 1) for n in range(6, 15): for m in range(1, n): if math.gcd(m, n) == 1: p.dot(p.X(m / n), p.Y(1 / n), SKY, 1.7) e_pts = [(0, 1.0), (1, 1.0)] for n in range(2, 6): for m in range(1, n): if math.gcd(m, n) == 1: e_pts.append((m / n, 1 / n)) for x, y in e_pts: p.hole(x, y, ORANGE, 4.4) p.dot(p.X(x), p.Y(y), BLUE, 2.6) p.xtick(0.5, '1/2', size=9.5) p.xtick(1.0, '1', size=9.5) p.ytick(0.2, '1/5', ORANGE, size=9.5) p.ytick(0.5, '1/2', size=9.5) p.ytick(1.0, '1', size=9.5) p.lab(0.5, 0.92, '11 circled points in E = {h(x) ≥ 1/5}', ORANGE, 9.8) p.lab(0.5, 0.08, 'outside E: h(x) < ε/2 = 1/5', GREEN, 9.5) return p.svg()
def F10_sgn_indefinite(): p1 = Plot(-1.15, 1.15, -1.35, 1.35, 360, 120, (26, 8, 14, 16)) p1.axes(xlab='', ylab='f') p1.seg(-1, -1, 0, -1, BLUE, 2.4) p1.seg(0, 1, 1, 1, BLUE, 2.4) p1.hole(0, -1, BLUE, 3.0) p1.hole(0, 1, BLUE, 3.0) p1.dot(p1.X(0), p1.Y(0), BLUE, 3.0) p1.ytick(1, '1', size=9) p1.ytick(-1, '−1', size=9) p1.lab(-0.95, 1.05, 'f(x) = sgn x (jumps at 0)', BLUE, 9.8, 'start') p2 = Plot(-1.15, 1.15, -1.25, 1.25, 360, 130, (26, 8, 14, 20)) p2.axes(xlab='x', ylab='F') p2.curve(abs, -1, 1, PURPLE, 2.4) p2.curve(lambda x: abs(x) - 1, -1, 1, ORANGE, 1.8, '4 3') p2.dot(p2.X(0), p2.Y(0), PINK, 3.4) p2.dot(p2.X(0), p2.Y(-1), PINK, 3.4) p2.xtick(-1, '−1', size=9) p2.xtick(1, '1', size=9) p2.lab(-0.62, 0.72, 'slope −1', PURPLE, 9.5) p2.lab(0.62, 0.72, 'slope +1', PURPLE, 9.5) p2.lab(0.08, -0.92, 'F(x) = |x| − 1 (corner at 0)', ORANGE, 9.5, 'start') return stack([(p1, 0, 0), (p2, 0, 120)], 360, 250)
def F11_area_sliver(): p = Plot(-0.1, 2.1, -0.2, 2.1, 360, 195, (26, 12, 14, 24)) f = lambda t: t c, h, eps = 1.0, 0.35, 0.25 p.axes(xlab='t', ylab='f(t)') p.hband(f(c) - eps, f(c) + eps, ORANGE, 0.14, 0, 2.0) p.area(f, c, c + h, AMBER, 0.42) p.box(c, c + h, f(c), GREEN, 0.18, GREEN) p.box(c, c + h, f(c) + eps, ORANGE, 0.12, ORANGE) p.curve(f, 0, 2.0, BLUE, 2.4) p.xtick(c, 'c', size=9.5) p.xtick(c + h, 'c+h', size=9.5) p.ytick(f(c), 'f(c)', size=9.5) p.lab(0.08, 1.82, 'sliver area = F(c+h) − F(c)', AMBER, 10, 'start') p.lab(0.08, 1.54, '(f(c)−ε)h ≤ ∫ f ≤ (f(c)+ε)h', ORANGE, 9.8, 'start') return p.svg()
def F12_substitution_panels(): p1 = Plot(-0.2, 4.3, -0.2, 2.15, 178, 190, (22, 12, 8, 22)) g = lambda t: math.sin(math.sqrt(t)) / math.sqrt(t) if t > 1e-4 else 1.0 p1.axes(xlab='t', ylab='y') p1.area(g, 1, 4, AMBER, 0.34) p1.curve(g, 0.01, 1.0, PINK, 1.6, '3 3') p1.curve(g, 1.0, 4.0, BLUE, 2.3) p1.xtick(1, '1', size=9) p1.xtick(4, '4', size=9) p1.ytick(1, '1', size=9) p1.lab(0.15, 1.92, 'sin√t / √t on [1,4]', BLUE, 9.5, 'start') p1.lab(1.2, 0.32, 'area 1.913', AMBER, 9.2, 'start')
p2 = Plot(-0.1, 2.3, -0.2, 2.15, 178, 190, (22, 12, 10, 22)) f = lambda x: 2 * math.sin(x) p2.axes(xlab='x', ylab='y') p2.area(f, 1, 2, AMBER, 0.34) p2.curve(f, 0, 2.2, PURPLE, 2.3) p2.xtick(1, '1', size=9) p2.xtick(2, '2', size=9) p2.ytick(2, '2', size=9) p2.lab(0.1, 1.92, '2 sin x on
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[1,2] (x=√t)', PURPLE, 9.5, 'start') p2.lab(1.05, 0.55, 'area 1.913', AMBER, 9.2, 'start') return stack([(p1, 0, 0), (p2, 182, 0)], 360, 190)
def F13_additivity_split(): f = lambda x: x * x p1 = Plot(-0.1, 2.15, -0.4, 4.3, 176, 190, (22, 12, 8, 22)) p1.axes(xlab='x', ylab='y') p1.area(f, 0, 1, SKY, 0.38) p1.area(f, 1, 2, AMBER, 0.38) p1.vline(1, INK, 1.4, '3 2', 0, 1) p1.curve(f, 0, 2, BLUE, 2.2) p1.xtick(1, 'c=1', size=9) p1.xtick(2, '2', size=9) p1.lab(0.08, 3.8, '∫₀¹ = 1/3', BLUE, 9.5, 'start') p1.lab(0.08, 3.1, '∫₁² = 7/3', ORANGE, 9.5, 'start')
p2 = Plot(-0.1, 2.15, -0.4, 4.3, 178, 190, (20, 12, 10, 22)) p2.axes(xlab='x', ylab='') p2.box(0, 0.8, f(0.5), SKY, 0.25, BLUE) p2.box(0.8, 1.0, f(1.0), AMBER, 0.40, ORANGE) p2.box(1.0, 1.3, f(1.0), AMBER, 0.40, ORANGE) p2.box(1.3, 2.0, f(1.6), SKY, 0.25, BLUE) p2.vline(1.0, PINK, 1.6, '3 2', 0, 2.2) p2.curve(f, 0, 2, BLUE, 2.0) p2.xtick(0.8, 'x□₋₁', size=8.5) p2.xtick(1.0, 'c', PINK, size=9) p2.xtick(1.3, 'x□', size=8.5) p2.lab(0.06, 3.8, 'Case (ii): split I□ at c', PINK, 9.2, 'start') p2.lab(0.06, 3.1, 'error ≤ 2M‖Q‖', ORANGE, 9.2, 'start') return stack([(p1, 0, 0), (p2, 182, 0)], 360, 190)
def F14_taylor_remainder(): p = Plot(-0.08, 1.18, -0.2, 3.0, 360, 200, (26, 12, 14, 24)) p.axes() t1 = lambda x: 1 + x t2 = lambda x: 1 + x + 0.5 * x * x p.curve(t1, 0, 1, GREEN, 1.8, '4 3') p.curve(t2, 0, 1, PURPLE, 2.0, '2 2') p.curve(math.exp, 0, 1, BLUE, 2.4) p.seg(1.0, 2.0, 1.0, math.e, ORANGE, 2.6) p.seg(1.05, 2.5, 1.05, math.e, PINK, 2.6) p.dot(p.X(1), p.Y(math.e), BLUE, 3.0) p.dot(p.X(1), p.Y(2.5), PURPLE, 2.8) p.dot(p.X(1), p.Y(2.0), GREEN, 2.8) p.xtick(1, '1', size=9.5) p.ytick(1, '1', size=9.5) p.ytick(math.e, 'e', BLUE, size=9.5) p.lab(0.05, 2.72, 'f(x) = eˣ', BLUE, 10, 'start') p.lab(0.05, 2.35, f'T{sub(1)}(x) = 1+x (R{sub(1)} = e−2)', GREEN, 9.5, 'start') p.lab(0.05, 1.98, f'T{sub(2)}(x) = 1+x+x²/2 (R{sub(2)} = e−5/2)', PURPLE, 9.5, 'start') return p.svg()
def F15_step_jump_strip(): p = Plot(-0.15, 3.15, -0.35, 3.6, 360, 195, (24, 12, 14, 24)) p.axes() d = 0.7 p.vband(1 - d, 1 + d, GREEN, 0.16, 0, 3.3) p.vline(1 - d, GREEN, 1.2, '3 3', 0, 3.3) p.vline(1 + d, GREEN, 1.2, '3 3', 0, 3.3) p.box(0, 0.7, 2, SKY, 0.28, BLUE) p.box(0.7, 1.3, 2, SKY, 0.28, BLUE) p.box(1.3, 1.8, 3, AMBER, 0.28, ORANGE) p.box(1.8, 2.4, 3, AMBER, 0.28, ORANGE) p.box(2.4, 3.0, 3, AMBER, 0.28, ORANGE) p.seg(0, 2, 1, 2, BLUE, 2.6) p.seg(1, 3, 3, 3, BLUE, 2.6) p.dot(p.X(1), p.Y(2), BLUE, 3.2) p.hole(1, 3, BLUE, 3.2) p.xtick(1 - d, '1−δ', GREEN, size=9) p.xtick(1, '1', size=9.5) p.xtick(1 + d, '1+δ', GREEN, size=9) p.xtick(3, '3', size=9.5) p.ytick(2, '2', size=9.5) p.ytick(3, '3', size=9.5) p.lab(1.55, 3.35, 'jump strip [1−δ, 1+δ]: |S − 8| ≤ 5δ', GREEN, 9.8, 'start') return p.svg()
def _step_piece(a, b, h, col, title, area_txt): p = Plot(-0.2, 3.2, -0.4, 3.6, 116, 170, (18, 10, 8, 20)) p.axes(xlab='', ylab='') p.box(a, b, h, col, 0.34, col) p.seg(a, h, b, h, col, 2.4) p.xtick(1, '1', size=8.5) p.xtick(3, '3', size=8.5) p.lab(0.1, 3.3, title, col, 9.2, 'start') p.lab(0.1, 2.65, area_txt, INK, 9.0, 'start') return p
def F16_elementary_steps(): p1 = step_piece(0, 1, 2, BLUE, '2φ{[0,1]}', 'area = 2·1 = 2') p2 = step_piece(1, 3, 3, ORANGE, '3φ{(1,3]}', 'area = 3·2 = 6') p3 = Plot(-0.2, 3.2, -0.4, 3.6, 118, 170, (18, 10, 8, 20)) p3.axes(xlab='', ylab='') p3.box(0, 1, 2, BLUE, 0.34, BLUE) p3.box(1, 3, 3, ORANGE, 0.34, ORANGE) p3.seg(0, 2, 1, 2, BLUE, 2.4) p3.seg(1, 3, 3, 3, ORANGE, 2.4) p3.xtick(1, '1', size=8.5) p3.xtick(3, '3', size=8.5) p3.lab(0.1, 3.3, 'g = sum', INK, 9.2, 'start') p3.lab(0.1, 2.65, '∫₀³ g = 8', GREEN, 9.2, 'start') return stack([(p1, 0, 0), (p2, 120, 0), (p3, 240, 0)], 360, 170)
def F17_rationals_cover(): p = Plot(-0.15, 1.15, -0.5, 1.2, 360, 155, (18, 12, 14, 18)) p.seg(0, 0, 1, 0, INK, 2.2) rats = [(0.0, 0.12, f'r{sub(1)}=0', 'ε/2', 0.35), (1.0, 0.06, f'r{sub(2)}=1', 'ε/4', 0.35), (0.5, 0.03, f'r{sub(3)}=1/2', 'ε/8', 0.35), (1 / 3, 0.015, f'r{sub(4)}=1/3', 'ε/16', 0.68)] for r, hw, rlab, wlab, y_iv in rats: p.vband(r - hw, r + hw, ORANGE, 0.24, -0.08, y_iv) p.seg(r - hw, y_iv, r + hw, y_iv, ORANGE, 2.0) p.dot(p.X(r), p.Y(0), BLUE, 3.0) p.lab(r, -0.28, rlab, BLUE, 9.2) p.lab(r, y_iv
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+ 0.14, wlab, ORANGE, 9.0) p.lab(0.5, 1.02, 'Total cover length = ε/2 + ε/4 + ε/8 + ⋯ = ε', INK, 10) return p.svg()
def F18_lipschitz_bound(): p = Plot(-0.1, 2.1, -0.25, 2.25, 360, 190, (26, 12, 14, 24)) f = lambda t: t w, z, M = 0.5, 1.5, 2.0 p.axes(xlab='t', ylab='f(t)') p.box(w, z, M, ORANGE, 0.16, ORANGE) p.area(f, w, z, AMBER, 0.42) p.hline(M, ORANGE, 1.2, '4 3', 0, 2.0) p.curve(f, 0, 2.0, BLUE, 2.4) p.xtick(w, 'w', size=9.5) p.xtick(z, 'z', size=9.5) p.ytick(M, 'M=2', ORANGE, size=9.5) p.brace_x(w, z, -0.14, 'z − w', GREEN, dy=12, size=9.5) p.lab(0.08, 1.72, 'F(z) − F(w) = ∫_w^z f', AMBER, 10, 'start') p.lab(0.08, 1.42, '|F(z) − F(w)| ≤ M|z − w|', ORANGE, 10, 'start') return p.svg()
=== CLOSING LIST OF NOTES AND DOUBTS (ALL 6 STAGES COMPLETE) ===

- All 7 files across the 6 stages have been output: src/m3/prepare_01_start.html, src/m3/prepare_02_sec71.html, src/m3/prepare_03_sec72.html, src/m3/prepare_04_sec73.html, src/m3/paper.html, src/m3/answers_1.html, src/m3/answers_2.html, src/m3/answers_3.html, and src/m3/figs_m3.py.
- Return type of functions in src/m3/figs_m3.py: Following the prompt rule "Each function returns an SVG string" and "In the HTML, place each figure as {{fig:F01_partition_norm}} with a one-line 'What to notice' caption beside it", each function in figs_m3.py returns <svg ...>...</svg> (p.svg() or stack(...)), and the HTML wraps <figure>{{fig:NAME}}<figcaption>...</figcaption></figure>.
- Bartle 7.3.18 proof misprint: Corrected (b-a)^{n-1} to (b-t)^{n-1} in the first displayed integral of the proof of Theorem 7.3.18 in both prepare_04_sec73.html (T32) and answers_3.html (Q36).
- Bartle 7.1.3 proof typo: Bartle prints \sum(f(x_i)-g(x_i))(x_i-x_{i-1}); we wrote the summand at the tags t_i and noted the book's print.
- PYQ count (74 vs 77): content_m3.md §5 lists 74 past-question tags, while pyq_all.md Part 8 contains 3 additional MTS6B10-22 descriptive rows (MTS-22 Q14, MTS-22 Q19, MTS-22 Q25). All 77 tags appear in src/m3/paper.html and the answer key.
- Ambiguous past questions noted in the answer key: MTS-22 MCQ 7 (countable exceptional set \{1/n\} vs finite set in Q6), MAT-22 MCQ 18 (both options A and B are true in Q6), MAT-22 Q18 (\int_0^4 \frac{\sin\sqrt t}{\sqrt t}\,dt evaluated by FTC 1 with E=\{0\} in Q30 since \varphi(t)=\sqrt t is not differentiable at 0), and MAT-22 Q33 (tagged 7.3.3/7.3.4 in the extract, treated as Theorem 7.3.5 in Q40).